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Figure 4.13: (a) Actinomorphic



Figure 4.13: (b) Zygomorphic



Figure 4.13: (c) Asymmetric

is an important structural adaptation related to pollination systems.

1. **Actinomorphic (or) radial or polysymmetric**: The flower shows two mirror images when cut in any plane or radius through the centre.Normally there are more than two planes of symmetry. Example: *Hibiscus, Datura*, water lily.

2. Zygomorphic (bilateral symmetry) or **monosymmetric**: The flower can be divided into equal halves in only one plane. Zygomorphic flower can efficiently transfer pollen grains to visiting pollinators. Example: *Pisum*, Bean, Cassia, Gulmohar, *Salvia, Ocimum*.

3. Asymmetric (amorphic): Flower lacks any plane of symmetry and cannot be divided into equal halves in any plane. Parts of such flowers are twisted. Example: *Canna indica*.

4.3 Accessory organs

4.3.1 Arrangement of whorls

The position of perianth (sepals, petals, tepals) parts relative to one another is called **perianth arrangement**.

1. **Cyclic or whorled:** All the floral parts are arranged in definite whorls. Example: *Brassica*,*Solanum.*

2. **Acyclic or spiral:** The floral parts are arranged in spirals on the elongated fleshy torus. Example: *Magnolia*.

3. **Spirocyclic or hemicyclic**: Some parts are in whorls and others parts are in spirals. Example: *Nymphaea, Annona, Polyalthia*

4.3.2 Cycly

It explains the number of whorls of floral parts. Perianth cycly is the number of whorls of perianth (sepals, petals) parts.



Figure 4.14: (a) Cyclic



Figure 4.14: (b) Acyclic



Figure 4.14: (c) Spirocyclic